NOT MEASUREMENT SENSITIVE

P-W-2891A
18 November 2002
SUPERSEDING
P-W-2891
31 July 1996

FEDERAL SPECIFICATION

WIPE SOLVENTS, EXEMPT AND NON-EXEMPT

INACTIVE FOR NEW DESIGN AFTER 19 APRIL 2001

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1. Scope. This specification covers exempt and non-exempt organic wipe solvents to be used in hand-wipe cleaning operations.

1.2 Classification.

1.2.1 Types. The types of wipe solvents are as follows:

Type I - Non-exempt, low vapor pressure composition (available only as classes B and C) (see 6.1).
Type II - Exempt composition (0 grams/liter volatile organic compound (VOC) content) (see 6.1).

1.2.1 Classes. The classes of wipe solvents are as follows:

Class A - Flash point of 20 to 99 ºF (-7 to +37 ºC)
Class B - Flash point of 100 to140 ºF (38 to 60 ºC)
Class C - Flash point above 140 ºF (60 ºC)
Class D - No flash point

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ 08733-5100.

FSC 7930

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2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FEDERAL SPECIFICATION

CCC-C-46	-	Cloth, Cleaning, Nonwoven Fabric. (Inactive for new design)

FEDERAL STANDARD

FED-STD-141	-	Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing

(Activities outside the Federal Government may obtain copies of federal specifications and standards as specified in the General Information section of the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index is for sale on a subscription basis from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification and other federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from the General Services Administration, Federal Supply Service, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC, 20407.)

(Federal Government activities may obtain copies of federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

MILITARY SPECIFICATIONS

MIL-DTL-5624	-	Turbine Fuel, Aviation, Grades JP-4, JP-5, and JP-5/JP-8 ST
MIL-PRF-23377	-	Primer Coatings: Epoxy, High-Solids
MIL-PRF-23699	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number O-156
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537
MIL-PRF-85582	-	Primer Coatings: Epoxy, Waterborne

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MILITARY STANDARDS

MIL-STD-129 - Standard Practice for Military Marking
MIL-STD-290 - Packaging and Marking of Petroleum and Related Products
MIL-STD-2073-1 - DoD Standard Practice for Military Packaging

(Copies of military specifications and standards required by contractors in connections with specific procurement functions are obtained from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

FEDERAL ACTS

DEPARTMENT OF ENVIRONMENTAL PROTECTION

Clean Air Act
Emergency Planning and Community Right-to-Know Act (EPCRA)
Superfund Amendments and Reauthorization Act (SARA)

(The Environmental Protection Agency acts are available from the Environmental Protection Agency, 1200 Pennsylvania Ave., Washington, DC 20406 or from their web site at http://www.epa.gov.)

FEDERAL REGULATIONS

DEPARTMENT OF ENVIRONMENTAL PROTECTION

40 CFR 51.100	-	Requirements for Preparation, Adoption, and Submittal of Implementation Plans: Definitions
40 CFR 63	-	National Emission Standards for Hazardous Air Pollutants for Source Categories: Aerospace Manufacturing and Rework Facilities
40 CFR 372	-	Toxic Chemical Release Reporting: Community Right-To-Know

DEPARTMENT OF TRANSPORTATION

49 CFR 171-178	-	Subchapter C - Hazardous Materials Regulations

(The Code of Federal Regulations (CFR) is for sale on a subscription basis from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the federal agency responsible for issuing them.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI-Z129.1	-	Chemicals, Hazardous Industrial, Precautionary Labeling of. (DoD adopted)

(Private sector and civil agencies may purchase copies of this voluntary standard from the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D56	-	Flash Point by Tag, Closed Cup Tester. (DoD adopted)
ASTM-D891	-	Gravity, Specific, Apparent, of Liquid Industrial Chemicals. (DoD adopted)
ASTM-D1296	-	Standard Test Method for Odor of Volatile Solvents and Diluents. (DoD adopted)
ASTM-D1353	-	Paint, Varnish, Lacquer and Related Products, Nonvolatile Matter in Volatile Solvents For Use In. (DoD adopted)
ASTM-D1364	-	Water in Volatile Solvents (Karl Fischer Reagent Titration Method). (DoD adopted)
ASTM-D1613	-	Acidity in Volatile Solvents and Chemical Intermediates Used in Paint, Varnish, Lacquer, and Related Products. (DoD adopted)
ASTM-D1876	-	Adhesives, Peel Resistance of, (T-Peel Test). (DoD adopted)
ASTM-D1901	-	Relative Evaporation Time of Halogenated Organic Solvents and Their Admixtures, Standard Test Method for
ASTM-D2109	-	Solvents and Their Admixtures, Nonvolatile Matter in Halogenated Organic. (DoD adopted)
ASTM-D2879	-	Vapor Pressure-Temperature Relationship and Initial Decomposition Temperature of Liquids by Isoteniscope, Test Method For
ASTM-D2942	-	Solvents, Halogenated Organic, Total Acid Acceptance Of (Nonreflux Methods). (DoD adopted)
ASTM-D2988	-	Water-soluble Halide Ion in Halogenated Organic Solvents and their Admixtures
ASTM-D2989	-	Solvents, Halogenated Organic and their Admixtures, Acidity-Alkalinity of. (DoD adopted)
ASTM-D3401	-	Water in Halogenated Organic Solvents and their Admixtures. (DoD adopted)
ASTM-D3545	-	Alcohol Content and Purity of Acetate Esters by Gas Chromatography. (DoD adopted)
ASTM-D3951	-	Packaging, Commercial. (DoD adopted)

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ASTM-D4755 - Solvents and their Admixtures, Halogenated Organic, Free Halogens in
ASTM-E299 - Solvents, Organic, Trace Amounts of Peroxides in
ASTM-F22 - Hydrophobic Surface Films by the Water-Break Test, Test Method for. (DoD adopted)
ASTM-F483 - Total Immersion Corrosion Test for Aircraft Maintenance Chemicals. (DoD adopted)
ASTM-F519 - Mechanical Hydrogen Embrittlement Evaluation of Plating Processes and Service Environments, Standard Test Method for. (DoD adopted)
ASTM-F945 - Stress-Corrosion of Titanium Alloys by Aircraft Engine Cleaning Materials, Standard Test Method for
ASTM-F1110 - Sandwich Corrosion Test, Standard Test Method for. (DoD adopted)
ASTM-F1111 - Corrosion of Low-Embrittling Cadmium Plate by Aircraft Maintenance Chemicals

(Private sector and civil agencies may purchase copies of these voluntary standards from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 704 - Identification System for Fire Hazards of Materials

(Private sector and civil agencies may purchase copies of this voluntary standard from the National Fire Protection Association, One Batterymarch Park, Quincy, MA 02269-9101.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE-AMS-QQ-A-250/4 - Aluminum Alloy 2024, Plate and Sheet. (DoD adopted)
SAE-AMS-QQ-A-250/5 - Aluminum Alloy Alclad 2024, Plate and Sheet. (DoD adopted)
SAE-AMS-QQ-A-250/12 - Aluminum Alloy 7075, Plate and Sheet. (DoD adopted)
SAE-AMS-QQ-A-250/13 - Aluminum Alloy Alclad 7075, Plate and Sheet. (DoD adopted)
SAE-AMS1640 - Corrosion Removing Compound for Aircraft Surfaces
SAE-AMS3276 - Sealing Compound, Integral Fuel Tanks and General Purpose, Intermittent Use to 360 °F (182 °C). (DoD adopted)
SAE-AMS3819 - Cloths, Cleaning, For Aircraft Primary and Secondary Structural Surfaces. (DoD adopted)

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SAE-AMS4377	-	Sheet and Plate, Magnesium Alloy, 3.0AL-1.0ZN-0.20MN (AZ31B-H24), Cold Rolled, Partially Annealed. (DoD adopted)
SAE-AMS4911	-	Titanium Alloy, Sheet, Strip, and Plate, 6AL-4V, Annealed. (DoD adopted)
SAE-AMS5040	-	Steel, Sheet and Strip, 0.15 Carbon, Maximum Deep Forming Grade. (DoD adopted)

(Private sector and civil agencies may purchase copies of these voluntary standards from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), samples shall be subjected to first article inspection in accordance with 4.3.

3.2 Toxicity. The wipe solvents supplied under this specification shall have no adverse effect on the health of personnel when used for their intended purpose.

3.3 Material. The wipe solvents supplied under this specification shall be clear, stable, and homogenous liquids, free of haze, rust particles, and suspended matter.

3.3.1 Composition. The wipe solvents shall be of the same composition and specific gravity as that of the first article sample, when tested in accordance with ASTM-D3545 for composition (modified to apply to the solvent used in the material) and ASTM-D891 for specific gravity. The wipe solvents shall meet the composition requirements of hand-wipe cleaning operations in accordance with 40 CFR 63.

3.3.2 Prohibited materials. The wipe solvents shall not contain any substance listed in:

a. section 602 of the Clean Air Act as an ozone-depleting substance
b. section 112 of the Clean Air Act as a hazardous air pollutant
c. sections 302 and 313 of the Emergency Planning and Community Right-to Know Act (Title III of the Superfund Amendments and Reauthorization Act [SARA]) (40 CFR 372)

3.4 Stability (shelf-life). The wipe solvents shall be stored in the original, unopened container for not less than one year at a temperature of 90 $\pm$2 ºF (34 $\pm$1 ºC) without degrading, and shall meet all the requirements of this specification after storage.

3.5 Performance. The wipe solvents shall meet the performance requirements in table I. For type II solvents containing any halogen, the wipe solvents shall meet the requirements of tables I and II. Failure of any test or inspection requirement shall result in the rejection of that lot from which the sample came.

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TABLE I. Performance requirements and test methods.

Property	Requirement	Test paragraph	ASTM test method
VOC content Type I Type II	 List result 0 g/l	4.5.1	--
Nonvolatile residue, parts per million (maximum) 1/	200	--	D1353
Water, weight percent (maximum) 1/	0.08	--	D1364
Acidity, as percent acetic acid (maximum) 1/	0.02	--	D1613
Peroxides, parts per million (ppm) (maximum)	5 ppm	--	E299
Odor	Not obnoxious	--	D1296
Relative evaporation time vs. reference solvent (minimum) 2/ Type I, Classes B and C Type II, Class A Type II, Classes B, C, and D	 0.2 2 0.8	--	D1901
Vapor pressure (type I only), millimeters (mm) of mercury at 20 °C (maximum)	7	--	D2879
Flash point, degrees Fahrenheit (°F) Class A Class B Class C Class D	 20 - 99 100 - 140 Above 140 No flash	--	D56
Flammability (class D only)	Rating of 1 or less	4.5.2	--
Sandwich corrosion	Rating of 1, or no effect greater than distilled water	--	F1110
Immersion corrosion weight change mg/cm^2/day (maximum) Aluminum, Titanium, Steel, and Cadmium Plated Steel Magnesium	 0.03 0.20	4.5.3	F483
Hydrogen embrittlement	No fracture of specimens in less than 200 hours at 75% of notch fracture strength (for alternate method, no fracture at less than 85% of notch fracture strength)	4.5.4	F519, type 1a
Titanium stress corrosion (examined with 500X magnification)	No cracking	--	F945, method A
Cleaning efficiency Lubricating oil (MIL-PRF-23699) Hydraulic fluid (MIL-PRF-83282) JP-5 jet fuel (MIL-DTL-5624)	 Water-break free Water-break free Water-break free	4.5.5	--
Adhesion Primer wet tape test Sealant T-peel strength, lb/in. (minimum)	 No primer peel or damage 10	4.5.6	--

1/ For halogen-containing type II materials, use the method and requirement listed in table II.
2/ The solvent evaporation time relative to the evaporation time of n-butylacetate shall be determined in accordance with ASTM-D1901, except n-butylacetate shall be substituted as the comparative reagent in the ASTM method.

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TABLE II. Performance requirements and test methods for type II solvents containing halogen.

Property	Requirement	Test paragraph	ASTM test method
Nonvolatile residue, parts per million (maximum)	50	---	D2109
Water, percent by weight (maximum)	0.015	---	D3401
Acidity, percent by weight (maximum)	0.001	---	D2989
Acid acceptance, percent by weight as NaOH (maximum)	0.16	---	D2942
Alkalinity, percent by weight as NaOH (maximum)	0.02	---	D2989
Halide, percent by weight (maximum)	0.0005	---	D2988
Free halogen	Negative	---	D4755

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. First article testing shall consist of all requirements and tests specified in this specification, and shall be accomplished on the first production lot of a contract or purchase order, when specified (see 6.2). Failure of any examination or test shall result in first article disapproval.

4.4 Quality conformance inspection. The quality conformance inspection shall consist of the following tests: composition, specific gravity, VOC content, non-volatile content, water content, acidity, acid acceptance, halide, odor, flash point, sandwich corrosion, and immersion corrosion. Any failure shall constitute cause for rejection.

4.4.1 Lot formation. A lot shall consist of all material manufactured at one time from one batch, forming part of one contract, and submitted for acceptance.

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4.5 Test methods. All tests performed for this specification shall be conducted in accordance with table I, table II, and 4.5.1 through 4.5.6.2.4. Unless otherwise stated in the test method, room temperature shall be 70 ±10 °F (21 ±5 °C) and relative humidity shall be 50 ±10 percent.

4.5.1 VOC content. The VOC content of type I wipe solvent shall be determined from the composition, specific gravity, and the following equation:

VOC content (g/l) = Specific gravity x 998.23 x (1- weight fraction exempt solvent)

where exempt solvents are those listed in 40 CFR 51.100. Each component of type II wipe solvent shall be verified as exempt by identifying its listing in 40 CFR 51.100.

4.5.2 Flammability (class D only). Flammability shall be determined in accordance with NFPA 704.

4.5.3 Immersion corrosion. The immersion corrosion test shall be conducted in accordance with ASTM-F483 on test panels constructed of the following materials:

a. Aluminum alloy 2024 (T3 temper), conforming to SAE-AMS-QQ-A-250/4
b. Aluminum alloy 7075 (T6 temper), conforming to SAE-AMS-QQ-A-250/12
c. Titanium alloy (6AL-4V), conforming to SAE-AMS4911
d. Carbon steel (1020), conforming to SAE-AMS5040
e. Cadmium plating applied to (d), in accordance with ASTM-F1111
f. Magnesium alloy (AZ31B-H24), conforming to SAE-AMS4377

4.5.4 Hydrogen embrittlement.

4.5.4.1 Specimens. Hydrogen embrittlement characteristics shall be determined in accordance with ASTM-F519, using cadmium-plated AISI 4340, type 1a specimens. Plating shall be applied using treatment B without conversion coating and shall cover the notch and surfaces within 0.5 inch (13 mm) of the notch; threaded surfaces shall not be plated. Plated specimens shall be baked in accordance with ASTM-F519.

4.5.4.2 Procedure. Four specimens shall be tested as follows. Stress the specimen by applying a load equivalent to 75 percent of notch fracture strength. Apply the solvent by brushing onto the notch area every 30 minutes for the first eight hours. Failure is as specified in table I. Alternatively (for step load testing), a load equivalent to 45 percent of notch fracture strength shall be applied for the first 24 hours. Apply the solvent by brushing onto the notch area every 30 minutes for the first eight hours. After the first 24 hours, step the load up an additional 5 percent each hour until failure. Failure is as specified in table I.

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4.5.5 Cleaning efficiency.

4.5.5.1 Test panel preparation. Prepare nine test panels, constructed from clad aluminum alloy 2024 (T3 temper) conforming to SAE-AMS-QQ-A-250/5, with dimensions of 6 by 3 by 0.025 inch (150 by 76 by 0.6 mm). Deoxidize the test panels by immersion for two minutes in corrosion removing compound conforming to SAE-AMS1640, type I, class 1. Upon removal, rinse the test panels with deionized water to obtain a water-break free surface. Allow test panels to air dry.

4.5.5.2 Test panel soiling. Each of three sets of three test panels shall be soiled with a different contaminant:

a. Turbine engine lubricating oil conforming to MIL-PRF-23699
b. Hydraulic fluid conforming to MIL-PRF-83282
c. JP-5 jet fuel conforming to MIL-DTL-5624.

Soil the test panels by evenly spreading 5 milliliters (mL) of each contaminant onto the surface of the test panel to be cleaned and coated with primer coating. Store the lubricating oil and hydraulic fluid contaminated test panels for 24 hours at 120 ±2 °F (49 ±1 °C). Air dry the jet fuel contaminated test panels 24 hours at room temperature.

4.5.5.3 Procedure. Clean the nine test panels as follows:

a. Fold a wipe cloth, conforming to CCC-C-46, type I, class 7, or SAE-AMS3819, grade A, and initial dimensions of 9 by 9 in (228 by 228 mm), into a square of 4.5 by 4.5 in (114 mm by 114 mm).
b. Soak the folded cloth with no more than 10 mL of test solvent.
c. Wipe each test panel with one stroke in one direction, applying moderate manual pressure. Ensure that the contact area between the cloth and the test panel is saturated with the test solvent. This procedure (a through c) shall not be repeated more than three times.
d. Allow to air dry for no more than one hour at ambient conditions (see 4.5).
e. Examine each test panel in accordance with ASTM-F22 for conformance to table I.

4.5.6 Adhesion.

4.5.6.1 Wet tape.

4.5.6.1.1 Test panel preparation. Prepare eighteen test panels, constructed from clad aluminum alloy 2024 (T3 temper) conforming to SAE-AMS-QQ-A-250/5, with dimensions of 6 by 3 by 0.025 in (150 by 76 by 0.6 mm). Deoxidize the test panels by immersion for two minutes in corrosion removing compound conforming to SAE-AMS1640, type I, class 1. Upon removal, rinse with deionized water to obtain a water-break free surface. Allow test panels to air dry.

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4.5.6.1.2 Test panel soiling. Each of three sets of six test panels shall be soiled with a different contaminant:

a. Turbine engine lubricating oil conforming to MIL-PRF-23699
b. Hydraulic fluid conforming to MIL-PRF-83282
c. JP-5 jet fuel conforming to MIL-DTL-5624.

Soil the test panels by evenly spreading five milliliters (mL) of contaminant onto the test surface of the test panel. Store the lubricating oil and hydraulic fluid contaminated test panels for 24 hours at 120 ±2 °F (49 ±1°C). Air dry the jet fuel contaminated test panels for 24 hours at room temperature.

4.5.6.1.3 Test panel cleaning. Lightly dampen a wipe cloth conforming to SAE-AMS3819, grade A, or CCC-C-46, class 7, with the test solvent and clean the soiled test panels with one stroke in one direction, applying moderate manual pressure. This procedure may be repeated no more than three times, using a clean wipe cloth and changing stroke direction each time. Allow the test panels to air dry.

4.5.6.1.4 Application of primer coating. Spray primer coating conforming to MIL-PRF-23377, type I, class C, to test panels for each contaminant within one hour of cleaning with the test solvent. Spray apply primer coating conforming to MIL-PRF-85582, type I, class C1 or N to the remaining test panels within one hour of cleaning with the test solvent. Store the coated test panels at 72 ±4 °F (22 ±2 °C) for not less than 14 days.

4.5.6.1.5 Procedure. Test the coated test panels in distilled water for 24 hours at room temperature, in accordance with FED-STD-141, method 6301. Examine the primer coatings for conformance to the requirements of table I.

4.5.6.2 T-peel.

4.5.6.2.1 Test panel preparation. Prepare twelve test panels, constructed from clad aluminum alloy 2024 (T3 temper) conforming to SAE-AMS-QQ-A-250/5, with dimensions of 12 by 6 by 0.016 in (304.8 by 152.4 by 0.41 mm). Deoxidize the test panels by immersing each for two minutes in corrosion removing compound conforming to SAE-AMS1640, type I, class 1. Upon removal, rinse with deionized water to obtain a water-break free surface. Allow test panels to air dry.

4.5.6.2.2 Test panel soiling. Soil twelve test panels, four test panels for each contaminant:

a. Hydraulic fluid conforming to MIL-PRF-83282
b. Turbine engine lubricating oil conforming to MIL-PRF-23699
c. JP-5 jet fuel conforming to MIL-DTL-5624.

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Soil the test panels by evenly spreading 5 mL of contaminant onto the test surface of each test panel. Store the lubricating oil contaminated and hydraulic fluid contaminated test panels for 24 hours at 120 ±2 °F (49 ±1 °C). Air dry the jet fuel contaminated test panels for 24 hours at room temperature (see 4.5).

4.5.6.2.3 Test panel cleaning. Lightly dampen a wipe cloth conforming to SAE-AMS3819, grade A, or CCC-C-46, class 7, with the test solvent and clean the test panels with one stroke in one direction, applying moderate manual pressure. This procedure may be repeated no more than three times, using a clean wipe cloth each time, and changing stroke direction each time. Allow the test panels to air dry.

4.5.6.2.4 Procedure.

a. On six pairs of panels, two pairs for each contaminant, mask the upper 3 in (68 mm) of each test panel by applying polytetrafluoroethylene (PTFE) masking tape.
b. Apply sealant conforming to SAE-AMS3276, Class A-2, to all of the test panels over the lower 9 in (230 mm) of each test panel. Do not apply sealant to the upper 3 in (68.0 mm) of each test panel.
c. After application of the sealant, place one layer of 0.005 in (0.13 mm) thick scrim cloth on top of the sealant on one of the test panels (the scrim cloth will cover 3/4 of the surface panel). Do not cover the PTFE taped portion of the test panel.
d. On top of the first panel, place another test panel that has been soiled with the same contaminant, cleaned, and had sealant applied, ensuring that the PTFE taped portions of the two panels are initially in contact prior to joining the sealant covered portions of the panels.
e. With a roller weighing 4 lb (1.8 kilograms), and starting at the PTFE taped end of the panel, apply manual pressure and push the roller over the surface of the joined panels. Maintain a uniform bond line thickness of the scrim cloth.
f. Wipe off the excess sealant that is squeezed out.
g. While the sealant between the test panels is curing, maintain a uniform pressure of not less than 34 kilopascals (kPa) (5 psi) on the test panels to control thickness and promote uniform contact between the bonding surfaces.
h. Cure the test panels for not less than 14 days at 75 ±4 °F (24 ±2 °C) and 50 percent relative humidity.
i. Cut the two sets of bonded test panels into strips measuring 12 by 1 by 0.016 in (304.8 by 25.4 by 0.41 mm). To eliminate edge effects, cut the first strip 0.5 in (12.7 mm) from the edge of the 6 in (152.4 mm) wide test panels. A total of five 1 in (25.4 mm) wide strips and two 0.5 in (12.7 mm) wide strips will result. In accordance with ASTM-D1876, test the 1 in (25.4 mm) test strips of each contaminant for conformance to the requirements of table I.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). In the absence of detailed packaging guidance, military packaging shall be accomplished in accordance with MIL-STD-2073-1. Commercial packaging as specified in ASTM-D3951 shall be used when military packaging is not required.

5.2 Marking. All marking shall conform to MIL-STD-129 and MIL-STD-290. In addition, individual containers shall bear printed labels with the following:

- Federal Specification P-W-2891, Wipe solvents, Exempt and Non-Exempt
- Type and Class
- Manufacturer's name and product number
- Date of manufacture by month and year
- Batch number
- Net contents
- Flashpoint
- VOC content

All unit and intermediate packs of toxic and hazardous chemicals and materials shall be labeled in accordance with all applicable laws, statutes, regulations and ordinances, including federal, state, and municipal requirements. All unit containers (including those that serve as shipping containers) shall be marked in accordance with 49 CFR 171-178 and with the applicable precautionary information detailed in ANSI-Z129.1.

5.2.1 Class A and B marking. Class A and B wipe solvents shall bear the following special marking:

"WARNING: This material is not permitted to be used onboard United States Navy ships."

6. NOTES

INFORMATION FOR GUIDANCE ONLY. (This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. This material supplied under this specification is intended to be used for hand-wipe cleaning of most aluminum, titanium, steel, and magnesium aircraft, and other aircraft component surfaces prior to painting, sealing, and non-structural adhesive bonding. The hand-wipe solvents are not intended to be sprayed, misted, or atomized in any manner; the solvents are intended to be poured or squirted onto wipe cloths for hand-wipe cleaning applications. The solvents are military unique since they are proven to clean a surface to the extent that will provide superior adhesion of coatings and sealants required for high performance military aircraft.

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Class A and B wipe solvents are not approved for use onboard United States Navy ships. Type II solvent is excluded from the definition of VOC and is listed in the EPA's regulation 40 CFR 51.100. VOC exempt refers to organic compounds that do not participate in atmospheric photochemical reactivity. The VOC definition in the 40 CFR 51.100(s) is subject to change. To ensure the most up-to-date regulation, contact the local air pollution control agency.

6.2 Acquisition requirements. Acquisition documents must specify the following, as a minimum:

a. Title, number, and date of this specification, including any amendments.
b. Type and class of wipe solvent required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Whether first article inspection is required (see 3.1 and 4.3). When required, specify the laboratory to perform inspection and forward results to the Naval Air Systems Command, Code 4.3.4.1, Bldg. 2188, 48066 Shaw Rd. Unit 5, Patuxent River, MD 20670-1908.
e. Whether a certified test report is required for the quality conformance inspection (see 4.4.1)
f. Packaging level, if other than commercial packaging is required (see 5.1).
g. Address where MSDSs must be sent (see 6.4).
h. Size of containers (see 6.6).

6.3 Subject term (key word) listing.

Cadmium
Non-volatile residue
Prepaint
Preseal
Solvent
Wipe cleaner

6.4 Material Safety Data Sheet (MSDS). A Material Safety Data Sheet must be prepared and submitted in accordance with FED-STD-313; additionally, it must conform to 29 CFR 1910.1200. The 29 CFR 1910.1200 requires that the MSDS for each hazardous chemical used in an operation must be readily available to personnel using the material. Contracting officers will identify the activities requiring copies of the MSDS.

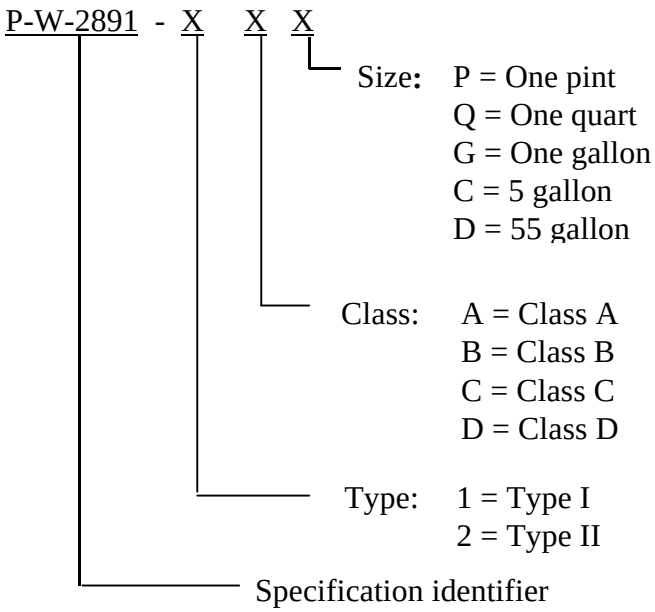
6.5 Relationship of this specification to previous issues. P-W-2891 initially provided two non-exempt wipe solvents under type I and type II and two classes of solvent with class B being compatible with acrylic and polycarbonate. This revision combines both non-exempt solvents under type I, eliminates the plastic compatibility classes, creates an exempt solvent under type II, and establishes classes corresponding to important flash point ranges. Cross-reference of

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classification changes between this specification (revision A) and the preceding issue (with no revision letter) are as follows:

P-W-2891A	P-W-2891
Type I, Class A	Not included
Type I, Class B	Type I, Class A
Type I, Class C	Type II, Class A
Type I, Class D	Not included
Type II, Class A	Not included
Type II, Class B	Not included
Type II, Class C	Not included
Type II, Class D	Not included
Not included	Type I, Class B
Not included	Type II, Class B

6.6 Part or identifying number (PIN). The PIN to be used for the wipe solvents acquired to this specification is as follows:



6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

Custodians:
Navy - AS
Air Force - 99

Review activity:
Air Force - 84

Civil coordinating agency:
GSA - FSS

Preparing activity:
Navy - AS
(DoD Project 7930-2440)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER P-W-2891A	2. DOCUMENT DATE (*YYYYMMDD*) 2002/11/18

3. DOCUMENT TITLE

WIPE SOLVENTS, EXEMPT AND NON-EXEMPT

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include ZIP Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	**7. DATE SUBMITTED** (*YYYYMMDD*)

8. PREPARING ACTIVITY

a. NAME COMMANDER NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION	b. TELEPHONE (*Include Area Code*) (1) Commercial (732) 323-2947 (2) DSN 624-2947
c. ADDRESS (*Include ZIP Code*) CODE 414100B120-3 HIGHWAY 547 LAKEHURST, NJ 08733-5100	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

DD Form 1426, FEB 1999 PREVIOUS EDITION IS OBSOLETE.

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